

Christian Giancola

MECHANICAL ENGINEERING | ROBOTIC CAM | ENGINEERING SOFTWARE

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PROFILE

Mechanical Engineering BSc candidate focused on robotic additive manufacturing, computational geometry and engineering software. I develop multi-axis toolpaths in Rhino and Grasshopper and Python postprocessors for KRL generation, coordinate transformations, tool orientation, extrusion control and external-axis coordination. I want to apply this combination of manufacturing knowledge and software development to automated CAD, CAM and robotic process-planning workflows.

EXPERIENCE

- Industrial Robotics and Additive Manufacturing Intern** | University of Bologna **2026 – Present**
Montecuccolino Laboratory, Bologna
- Developed software-driven toolpaths for robotic material deposition using a six-axis KUKA robot, Rhino 8, Grasshopper and KUKA|prc.
 - Designed planar, inclined and non-planar slicing strategies for complex surfaces and multi-axis trajectories.
 - Developed Python postprocessors that convert geometric path data into KRL programs and manage coordinate transformations, tool orientation, extrusion, an external linear axis, homing and purge sequences.
 - Integrated, configured and tested a system comprising the robot, extrusion unit, tool and external linear axis.
 - Built parametric geometry-processing workflows for curve ordering, path continuity, surface-normal control and local tool-frame generation.
 - Simulated and checked reachability, singularities, orientation and trajectories before execution on the robot.
 - Performed hands-on printing tests and iteratively modified geometry, process parameters and code based on observed results.

SELECTED PROJECTS

- Toolpath Check** | Personal software project **2026**
- Browser-based tool for analysing and validating robotic paths through geometric checks, program inspection and workflow automation.
- KinePath** | Personal software project **2026**
- Engineering application for preparing and managing industrial robot paths, developed from laboratory process-planning requirements.
- FrameForge** | Personal software project **2026**
- Browser-based platform for generating usable CAD geometry from images, videos and mechanical drawings.

EDUCATION

BSc in Mechanical Engineering | University of Bologna

2023 – Present

- Experimental thesis on the generation and control of toolpaths for robotic additive manufacturing.
- Main areas: mechanics, mechanical design, manufacturing technology, automation, programming and industrial systems.

Academic Diploma in Classical Guitar | Conservatory

2024 – Present

- Parallel programme requiring precision, discipline and long-term management of complex activities.

TECHNICAL SKILLS

Automation and robotics: KUKA, KRL, KUKA|prc, offline programming, simulation, six-axis robots, external axes

Robotic CAM: robotic deposition, planar and non-planar slicing, multi-axis toolpaths, tool orientation, collision-aware planning concepts and technical validation

Software development: Python, C++, postprocessing, geometric-data processing, browser-based engineering tools and workflow automation

Computational design: Rhino 8, Grasshopper, curves, surfaces, Breps, meshes, data trees and computational geometry

CAD and drafting: PTC Creo, AutoCAD, 3D modelling and technical drawings

Engineering methods: prototyping, machine testing, troubleshooting, technical documentation and iterative improvement

LANGUAGES

Italian: native • **English:** B2, Cambridge First Certificate, Grade C, used for technical documentation and communication

ADDITIONAL INFORMATION

- Italian and EU citizen, driving licence B.
- Available from February 2027 for a short technical placement or engineering software project in London.
- Basketball referee with experience making decisions and communicating under pressure.